

Lesson 4

Detection theory

1 Key concepts

Before doing these exercises it is important to review and understand the following concepts:

- General diagram of a digital communication system.
- Concepts of orthonormal basis, signal space and constellation.
- How an orthonormal basis is computed from a set of transmitted signals.
- Decision criteria (MAP and ML). Decision regions.
- Calculation of the probability of error.
- Bounds. Differences between the union bound and the simplified union bound.
- Demodulator implementation: bank of correlators or matched filters.
- Decisor implementation: theoretical scheme or practical implementation.

2 Basic problems

This first part includes problems mostly extracted from the bibliography aimed at practicing basic calculations needed for the rest of the lesson.

Problem 4.1

[Artés2012] Consider the constellation of signals $\{s_i(t), i = 0, \dots, M - 1\}$, identically null outside the interval $0 \leq t < T$, and of value

$$s_i(t) = \sin\left(\frac{2\pi(i+1)t}{T}\right)$$

inside it. Determine an orthonormal basis to represent the constellation of signals, its coordinates with respect to this basis, the distance between any two signals, the energy of each signal in the constellation, and the mean energy of the constellation.

RESULTS FOR PROBLEM

a) $\phi_i(t) = \sqrt{\frac{2}{T}} \sin\left(\frac{2\pi(i+1)t}{T}\right)$

b) $d_{ij} = \sqrt{T}$

c) $E_i = E_s = \frac{T}{2}$

Problem 4.2

[Sklar2001] Determine whether or not $s_1(t)$ and $s_2(t)$ are orthogonal over the interval $(-1.5T < t < 1.5T)$, where $s_1(t) = \cos(\omega_1 t + \phi_1)$, $s_2(t) = \cos(\omega_2 t + \phi_2)$, and $\omega_2 = 2\pi/T$ for the following cases:

a) $\omega_1 = \omega_2$ and $\phi_1 = \phi_2$

- b) $\omega_1 = 2\omega_2$ and $\phi_1 = \phi_2$
- c) $\omega_1 = \omega_2$ and $\phi_1 = \phi_2 + \pi/2$
- d) $\omega_1 = \omega_2$ and $\phi_1 = \phi_2 + \pi$

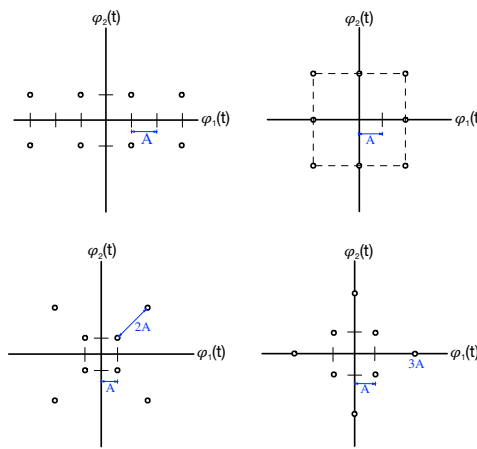
RESULTS FOR PROBLEM

- a) Not orthogonal
- b) Orthogonal
- c) Orthogonal
- d) Not orthogonal

Problem 4.3

Consider the constellations plotted below, where the signals are equiprobable. For each of them, calculate a. and b.:

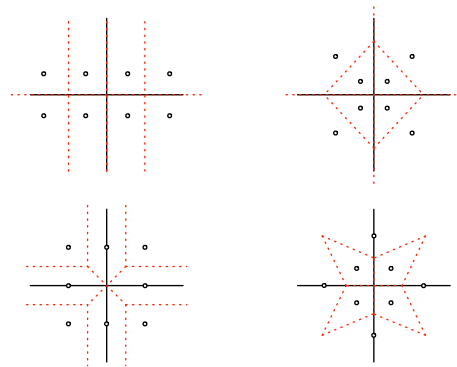
- The value of the constant A so that $E_s = 1J$.
- The decision regions and a bound on the probability of error for $N_0/2 = -13dBW/Hz$.



RESULTS FOR PROBLEM

a) $A = \frac{1}{\sqrt{6}}, A = \frac{1}{\sqrt{6}}, A = \frac{1}{\sqrt{2(2+\sqrt{2})}}, A = \sqrt{\frac{2}{11}}$

b) $P_e = 0.2515, P_e = 0.2515, P_e = 0.31199, P_e = 0.2010$



Problem 4.4

Consider the following signal set, defined within the interval $-1 \leq t < 1$.

$$s_1(t) = 1$$

$$s_2(t) = t$$

$$s_3(t) = t^2$$

Calculate:

- An orthonormal basis to represent said signal set, by using the Gram-Schmidt method.
- Plot the corresponding constellation.

RESULTS FOR PROBLEM

a) $\phi_1(t) = \frac{1}{\sqrt{2}}; -1 \leq t < 1$

$$\phi_2(t) = \sqrt{\frac{3}{2}}t; -1 \leq t < 1$$

$$\phi_3(t) = \sqrt{\frac{45}{8}}\left(t^2 - \frac{1}{3}\right), -1 \leq t < 1$$

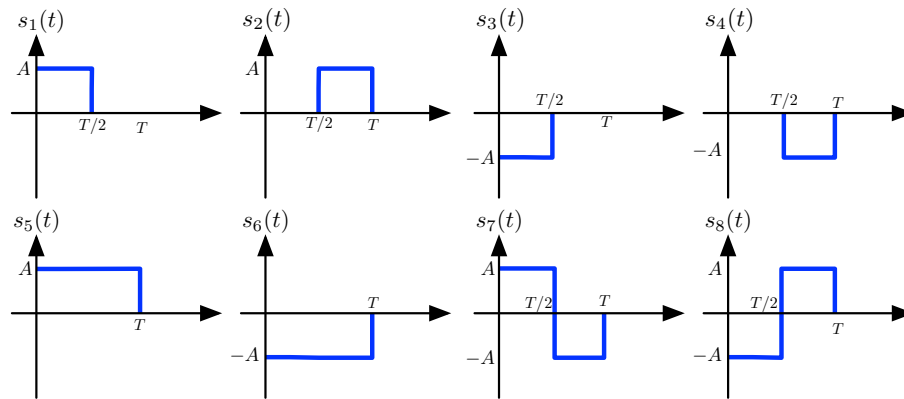
b) $s_1(t) = \sqrt{2} \cdot \phi_1(t)$

$$s_2(t) = \sqrt{\frac{2}{3}} \cdot \phi_2(t)$$

$$s_3(t) = \frac{\sqrt{2}}{3} \cdot \phi_1(t) + \sqrt{\frac{8}{45}} \cdot \phi_3(t)$$

Problem 4.5

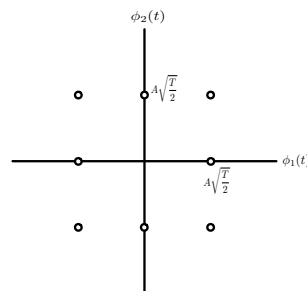
In order to transmit a set of eight equiprobable symbols by means of a baseband system, the eight signals in the figure are chosen.



- a) Depict a possible constellation for this signal set and give the distance between adjacent signals.

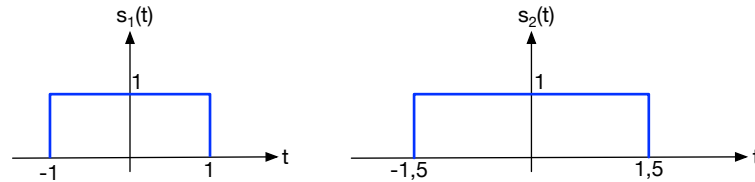
RESULTS FOR PROBLEM

a) $d_{12} = d_{14} = A \cdot \sqrt{T}$, $d_{13} = A \cdot \sqrt{2T}$, $d_{15} = d_{17} = A \cdot \sqrt{\frac{T}{2}}$, $d_{16} = d_{18} = A \cdot \sqrt{\frac{5T}{2}}$



Problem 4.6

The signals depicted below are used to transmit two possible equiprobable messages across a channel distorted by AWGN, whose power density spectrum is $N_0/2 = 1W/Hz$.

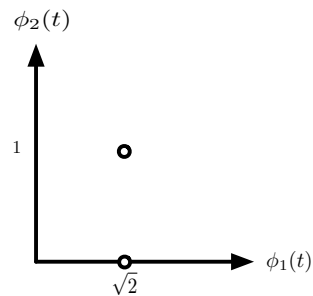


- Calculate and plot an orthonormal basis for this signal set.
- Plot the corresponding constellation.
- Calculate the corresponding error probability.

RESULTS FOR PROBLEM

- $$\phi_1(t) = \frac{1}{\sqrt{2}}s_1(t) \quad -1 \leq t < 1$$

$$\phi_2(t) = s_2(t) - s_1(t) \quad -1.5 \leq t < 1.5$$
- $$s_1(t) = \sqrt{2}\phi_1(t), \quad s_2(t) = \sqrt{2}\phi_1(t) + \phi_2(t)$$



- $$P_e = Q(0.5) = 0.30853$$

Problem 4.7

Consider the signal set given by:

$$s_i(t) = \begin{cases} \sqrt{\frac{2E}{T}} \cos(\omega_c t + i\frac{\pi}{4}) & 0 \leq t < T \\ 0 & \text{c.c.} \end{cases}$$

where $i = 1, 2, 3, 4$ and $\omega_c = n_c \cdot \frac{2\pi}{T}$ for some integer value n_c .

- Which is the dimension, N , of the space generated by this signal set?
- Obtain an orthonormal basis for this set.
- Obtain the signal vectors for each of the signals $s_i(t)$.
- Plot the signal vectors in the signal space.

RESULTS FOR PROBLEM

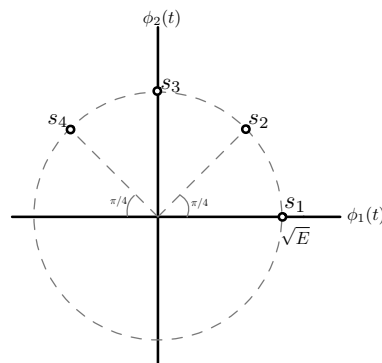
- $N = 2$
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$$\begin{aligned} \phi_1(t) &= \sqrt{\frac{2}{T}} \cos(\omega_c t + \frac{\pi}{4}) & 0 \leq t < T \\ \phi_2(t) &= \sqrt{\frac{2}{T}} \cos(\omega_c t + \frac{3\pi}{4}) & 0 \leq t < T \end{aligned}$$

-

$$\begin{aligned} \bar{s}_1 &= [\sqrt{E}, 0] & \bar{s}_2 &= \left[\sqrt{\frac{E}{2}}, \sqrt{\frac{E}{2}} \right] \\ \bar{s}_3 &= [0, \sqrt{E}] & \bar{s}_4 &= \left[-\sqrt{\frac{E}{2}}, \sqrt{\frac{E}{2}} \right] \end{aligned}$$

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Problem 4.8

A digital transmission system transmits the symbols 0 and A . At the receiver, the noise can be considered as white and Gaussian, with power spectral density $N_0/2 = 0.01W/Hz$. If we suppose rectangular pulses, calculate the amplitude at the transmitter to guarantee that only 1 out of 10^6 bits is wrongly received.

RESULTS FOR PROBLEM

$$A > 0.96$$

Problem 4.9

Consider a digital system with equiprobable digits 0 and A , and $\left(\frac{S}{N}\right)_R = 50$.

Calculate the error probabilities $P_{e|0}$ (error probability conditioned to the transmission of symbol 0), $P_{e|A}$ and P_e when the threshold is situated in $V = 0.4A$, and compare it when the minimum given by the equation $P_e = Q\left(\frac{A}{2\sigma}\right)$.

RESULTS FOR PROBLEM

- $P_{e|0} = Q(4)$
- $P_{e|1} = Q(6)$
- $P_e = 1.58 \cdot 10^{-5}$
- $P_{e_{min}} = 2.87 \cdot 10^{-7}$

3 Additional problems

These problems are slightly more elaborated than the previous ones, in many cases extracted from old exams.

Problem 4.10

Every T seconds, a transmitter sends across an AWGN channel one of the four following equiprobable waveforms.

$$\begin{aligned}s_1(t) &= \phi_1(t) \\ s_2(t) &= \phi_2(t) \\ s_3(t) &= \phi_3(t) \\ s_4(t) &= \phi_1(t) + \phi_2(t) + \phi_3(t)\end{aligned}$$

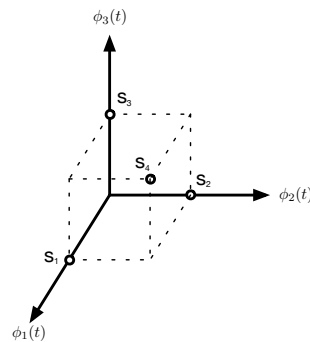
$$\phi_i(t) = \sqrt{\frac{2}{T}} \cos(\omega_i t)$$

$$0 \leq t < T \quad \omega_i = \frac{2\pi i}{T} \quad i = 1, 2, 3$$

- Explain whether or not the signals $\phi_i(t)$ constitute an orthonormal basis.
- Plot the constellation corresponding to this signal set.

RESULTS FOR PROBLEM

- Yes, the signals constitute an orthonormal basis.
-



Problem 4.11

A binary transmission system employs the signals $s_0(t)$ and $s_1(t)$ to transmit two equiprobable hypothesis, across a system affected by AWGN with power spectral density $N_0/2$.

$$\begin{aligned} s_0(t) &= A \cdot \cos\left(\frac{\pi t}{T}\right) & 0 \leq t < T \\ s_1(t) &= A \cdot \cos\left(\frac{\pi t}{T} + \phi\right) & 0 \leq t < T \end{aligned}$$

where T is the symbol period.

- Calculate an orthonormal basis for $s_0(t)$ and $s_1(t)$.
- Obtain the constellation corresponding to both signals and plot the decision regions.
- Calculate the error probability.

RESULTS FOR PROBLEM

a)

$$\begin{aligned} \phi_1(t) &= \sqrt{\frac{2}{T}} \cos\left(\frac{\pi t}{T}\right) & 0 \leq t < T \\ \phi_2(t) &= -\sqrt{\frac{2}{T}} \sin\left(\frac{\pi t}{T}\right) & 0 \leq t < T \end{aligned}$$

b)

$$\begin{aligned} s_0(t) &= A\sqrt{\frac{T}{2}} \cdot \phi_1(t) \\ s_1(t) &= A\sqrt{\frac{T}{2}} \cdot \cos(\phi) \cdot \phi_1(t) + A\sqrt{\frac{T}{2}} \cdot \sin(\phi) \cdot \phi_2(t) \end{aligned}$$

Decision regions:

- Z_1 (s_1 is decided): $\frac{\phi}{2} < \theta < (\frac{\phi}{2} + \pi)$
- Z_0 (s_0 is decided): $(\frac{\phi}{2} - \pi) \leq \theta \leq \frac{\phi}{2}$

c) $P_e = Q\left(A\sqrt{\frac{T}{N_0}} \sin\frac{\phi}{2}\right)$

Problem 4.12

[Sep1999] A communication system transmits the signals $s_i(t)$, $i = 0, 1, 2, 3, 4, 5, 6, 7$.

$$s_i(t) = A \cdot \cos\left(\omega_c t + \frac{\pi}{4}i\right); \quad 0 \leq t < T; \quad \omega_c = n_c \cdot \frac{2\pi}{T}; \quad n_c \in \mathbb{Z}$$

across an AWGN channel with power spectral density $N_0/2$.

- Provide an orthonormal basis to represent these signals.
- Plot the general block diagram of a correlator receiver designed to detect these signals.
- Calculate an upper bound for the symbol error probability when we have the data $A = 0.2mV$, $N_0/2 = 10^{-15}W/Hz$ y $T = 8\mu s$.

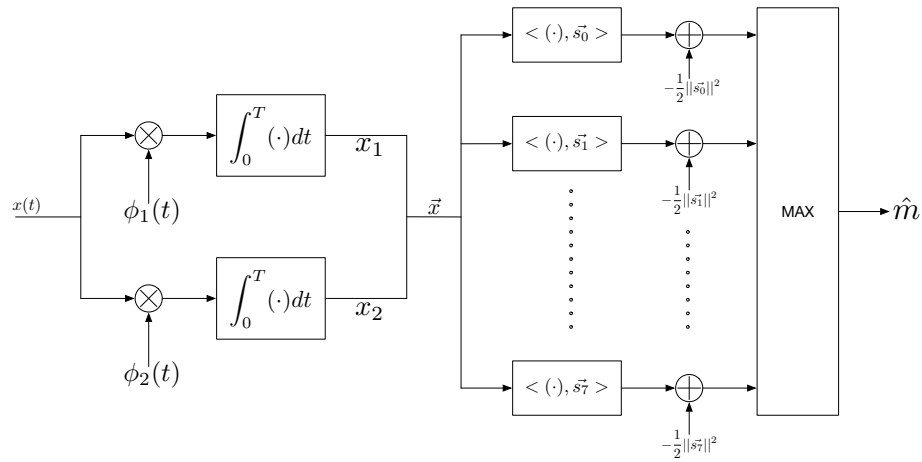
NOTE: $P_e \leq (M-1) \cdot Q\left(\frac{d_{min}}{\sqrt{2N_0}}\right)$

RESULTS FOR PROBLEM

- Orthonormal basis:

$$\phi_1(t) = \sqrt{\frac{2}{T}} \cdot \cos(\omega_c t) \quad \phi_2(t) = \sqrt{\frac{2}{T}} \cdot \sin(\omega_c t)$$

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- $P_e \leq 5.55 \cdot 10^{-6}$

Problem 4.13

[June1999] The signals $s_i(t)$, $i = 1, 2, 3, 4$, are used to transmit four equiprobable messages across an AWGN channel with power spectral density $N_0/2$.

$$s_1(t) = \begin{cases} \frac{A}{2} & 0 \leq t < \frac{T}{2} \\ -\frac{A}{2} & \frac{T}{2} \leq t < T \end{cases} \quad s_2(t) = \begin{cases} -\frac{A}{2} & 0 \leq t < \frac{T}{2} \\ \frac{A}{2} & \frac{T}{2} \leq t < T \end{cases}$$

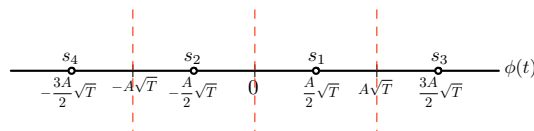
$$s_3(t) = \begin{cases} \frac{3A}{2} & 0 \leq t < \frac{T}{2} \\ -\frac{3A}{2} & \frac{T}{2} \leq t < T \end{cases} \quad s_4(t) = \begin{cases} -\frac{3A}{2} & 0 \leq t < \frac{T}{2} \\ \frac{3A}{2} & \frac{T}{2} \leq t < T \end{cases}$$

- Obtain an orthonormal basis for the signal space considered and give its dimension.
- Plot the signal constellation and the decision regions corresponding to the criterion of minimum error probability.
- Plot the impulse responses of the matched filters corresponding to each of the basis signals.
- Calculate the symbol error probability as a function of A and $N_0/2$.

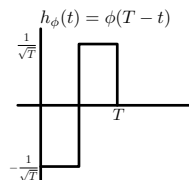
RESULTS FOR PROBLEM

a) $\phi(t) = \begin{cases} \frac{1}{\sqrt{T}} & 0 \leq t < \frac{T}{2} \\ -\frac{1}{\sqrt{T}} & \frac{T}{2} \leq t < T \end{cases}$

b)



c)



d) $P_e = \frac{3}{2} Q\left(A\sqrt{\frac{T}{2N_0}}\right)$

Problem 4.14

A digital communication system uses the following signals to transmit four equiprobable symbols:

$$\begin{aligned}s_1(t) &= 0 \\s_2(t) &= \text{sen}(\omega_0 t) \\s_3(t) &= \cos(\omega_0 t) \\s_4(t) &= \sqrt{2} \cdot \text{sen}\left(\omega_0 t + \frac{\pi}{4}\right)\end{aligned}$$

where $0 \leq t < T$ and $\omega_0 = n \cdot \frac{2\pi}{T}$ with $n \in \mathbb{Z}$.

- Obtain an orthonormal basis set to represent the given signals.
- Obtain the signal vectors for each of the signals $s_i(t)$ and plot the constellation.
- Calculate the simplified union bound of the probability of symbol error knowing that $N_0/2 = 10^{-11} \text{ W/Hz}$ and $T = 0.8 \text{ ns}$.

RESULTS FOR PROBLEM

a)

$$\begin{aligned}\phi_1(t) &= \sqrt{\frac{2}{T}} \cos(\omega_0 t) \quad 0 \leq t < T \\ \phi_2(t) &= \sqrt{\frac{2}{T}} \text{sen}(\omega_0 t) \quad 0 \leq t < T\end{aligned}$$

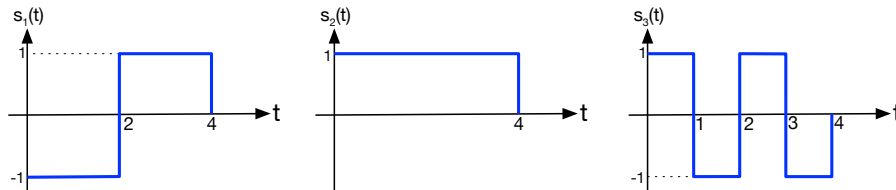
b)

$$\begin{aligned}\bar{s}_1 &= [0, 0] & \bar{s}_2 &= \left[0, \sqrt{\frac{T}{2}}\right] \\ \bar{s}_3 &= \left[\sqrt{\frac{T}{2}}, 0\right] & \bar{s}_4 &= \left[\sqrt{\frac{T}{2}}, \sqrt{\frac{T}{2}}\right]\end{aligned}$$

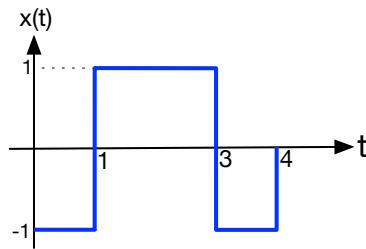
c) $P_e \leq 2.902 \cdot 10^{-3}$

Problem 4.15

The following signals are used to send three equiprobable symbols:



- Obtain an orthonormal basis set to represent the given signals.
- Plot the signals in the signal space.
- Provide a bound on the probability of error knowing that the AWGN channel is characterized by a power spectral density of $N_0/2 = 10^{-2}W/Hz$.
- Use the set of basis obtained in a. to represent the following signal $x(t)$ and plot it in the constellation.

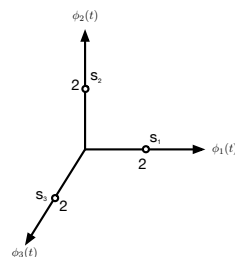


$$x(t) = \begin{cases} -1 & 0 \leq t < 1 \\ 1 & 1 \leq t < 3 \\ -1 & 3 \leq t < 4 \end{cases}$$

- Suppose that signal $x(t)$ is received. Justify which symbol would the receiver detect in this case.

RESULTS FOR PROBLEM

- $\phi_i(t) = \frac{1}{2} \cdot s_i(t) ; i = 1, 2, 3$
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- $P_e \leq 2 \cdot Q(14.14)$
- $\bar{x} = [0, 0, 0]$.
- $\bar{x}$ is equidistant from the three symbols: the receiver would randomly choose one of them.

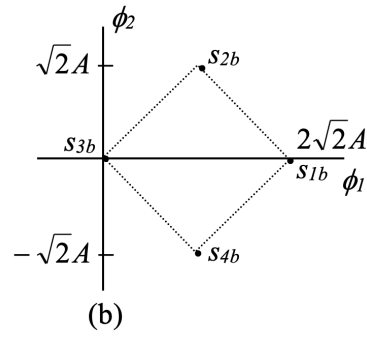
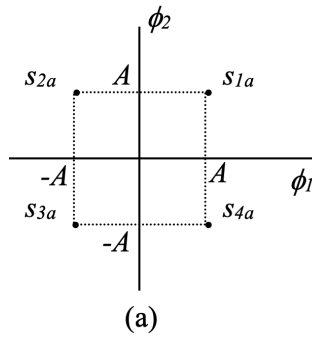
Problem 4.16

[June2001] Given the two constellations in the figure, where the base signals are:

$$\phi_1(t) = \begin{cases} \sqrt{\frac{2}{T}} \cos\left(\frac{2\pi}{T}t\right) & 0 \leq t < T \\ 0 & \text{c.c.} \end{cases}$$

$$\phi_2(t) = \begin{cases} \sqrt{\frac{2}{T}} \sin\left(\frac{2\pi}{T}t\right) & 0 \leq t < T \\ 0 & \text{c.c.} \end{cases}$$

- Obtain the temporal expressions of the signals of the two constellations.
- Obtain the average energy transmitted per symbol (E_s) in each of the two constellations.
- Compute the union bound of the symbol error probability as a function of E_s/N_0 in the two constellations.



RESULTS FOR PROBLEM

$$\begin{aligned} \text{a) } s_{1a}(t) &= A \cdot \phi_1(t) + A \cdot \phi_2(t) \\ s_{2a}(t) &= -A \cdot \phi_1(t) + A \cdot \phi_2(t) \\ s_{3a}(t) &= -A \cdot \phi_1(t) - A \cdot \phi_2(t) \\ s_{4a}(t) &= A \cdot \phi_1(t) - A \cdot \phi_2(t) \\ s_{1b}(t) &= 2\sqrt{2}A \cdot \phi_1(t) \\ s_{2b}(t) &= \sqrt{2}A \cdot \phi_1(t) + \sqrt{2}A \cdot \phi_2(t) \\ s_{3b}(t) &= 0 \\ s_{4b}(t) &= \sqrt{2}A \cdot \phi_1(t) - \sqrt{2}A \cdot \phi_2(t) \end{aligned}$$

$$\begin{aligned} \text{b) } E_{sa} &= 2A^2 \\ E_{sb} &= 4A^2 \end{aligned}$$

$$\text{c) } P_e \leq 2Q\left(\frac{\sqrt{2}A}{\sqrt{N_0}}\right) + Q\left(\frac{2A}{\sqrt{N_0}}\right)$$

Problem 4.17

[June2003] A digital communication system transmits every T seconds a signal chosen in an equiprobable way among the elements of the set $\{s_i(t)\}, i = 1, 2, 3$. It is requested:

- Get an orthonormal basis to represent $\{s_i(t)\}$.
- Calculate the energy of each of the signals.
- Draw the signal constellation, expressly indicating the coordinates of each signal point.
- Compute the simplified union bound of the error probability as a function of the mean symbol energy, E_s , for an additive Gaussian white noise channel with zero mean and power spectral density $N_0/2$ W/Hz when a correlation receiver is used.

$$s_1(t) = \begin{cases} 1 & 0 \leq t < T \\ 0 & \text{c.c.} \end{cases}$$

$$s_2(t) = \begin{cases} -\cos\left(\frac{\pi}{T}t\right) & 0 \leq t < T \\ 0 & \text{c.c.} \end{cases}$$

$$s_3(t) = \begin{cases} \cos^2\left(\frac{\pi}{2T}t\right) & 0 \leq t < T \\ 0 & \text{c.c.} \end{cases}$$

RESULTS FOR PROBLEM

- $\phi_1(t) = \frac{s_1(t)}{\sqrt{T}}$
 $\phi_2(t) = \sqrt{\frac{2}{T}} s_2(t)$
- $E_1 = T, E_2 = \frac{T}{2}, E_3 = \frac{3T}{8}$
 $E_s = \frac{5T}{8}$
- $\bar{s}_1 = (\sqrt{T}, 0)$
 $\bar{s}_2 = (0, \sqrt{\frac{T}{2}})$
 $\bar{s}_3 = (\frac{\sqrt{T}}{2}, -\frac{1}{2}\sqrt{\frac{T}{2}})$
- $P_e \leq 2Q\left(\sqrt{\frac{3E_s}{10N_0}}\right)$

References

[Artés2012] Antonio Artés et al. Comunicaciones Digitales. http://www.tsc.uc3m.es/antonio/libro_comunicaciones.

[Sklar2001] Bernard Sklar. Digital Communications 2nd Ed. Pearson, 2001.